

Air Quality in Delhi

An Analytical Study of Pollutant Dynamics, Diurnal Behaviour,
and Environmental Drivers of the Air Quality Index

Data period analysed: 1–24 January 2023 (hourly observations)

Source measurements: co, no, no₂, o₃, so₂, pm_{2.5}, pm₁₀, nh₃ (µg/m³)

Executive Summary

Delhi's air quality remains one of the most pressing environmental and public-health challenges facing any megacity in the world. This report analyses 561 hours of continuous pollutant monitoring (CO, NO, NO₂, O₃, SO₂, PM_{2.5}, PM₁₀, and NH₃) covering 1–24 January 2023, converts these measurements into the Central Pollution Control Board's (CPCB) National Air Quality Index framework, and examines the temporal patterns, pollutant relationships, and likely geographical drivers behind them.

The period examined was overwhelmingly dominated by the two most severe CPCB categories. The report finds that the AQI sat in the "Severe" band (>400) for 76% of hours and "Very Poor" (301–400) for a further 22%, with fine particulate matter (PM_{2.5}) acting as the AQI-determining pollutant in 90% of hours. PM_{2.5} and PM₁₀ concentrations exceeded the WHO 24-hour guideline (15 µg/m³) on every single day of the study, averaging roughly 24 times that threshold.

Two structural patterns stand out. First, a pronounced diurnal cycle: concentrations dip during midday convective mixing (12:00–15:00 IST) and rise sharply through the evening rush hour into an overnight peak, consistent with Delhi's shallow winter boundary layer and radiative surface cooling. Second, a statistically significant weekday–weekend gap ($p < 0.001$ for PM_{2.5}, NO₂, and CO), pointing to a substantial traffic and commercial-activity contribution layered on top of a persistent regional pollution baseline.

These findings reinforce that Delhi's winter air-quality crisis is not a single-pollutant or single-source problem: it reflects the interaction of vehicular and combustion emissions, regional agricultural and industrial contributions, and a geography and winter meteorology that combine to trap pollutants close to the surface. Section 8 translates these findings into targeted recommendations for monitoring, source control, and public health response.

1. Introduction and Environmental Context

Delhi, the National Capital Territory of India and one of the world's most populous urban agglomerations, has for over a decade recorded some of the highest ambient air pollution levels of any major city. Its air-quality crisis is shaped by a distinctive combination of geography, climate, and rapid urbanisation:

- **Landlocked, continental setting:** Delhi sits in the Indo-Gangetic Plain, far from any moderating coastline. Without sea breezes to ventilate the airshed, pollutants that build up over the city and surrounding states have limited natural means of dispersal.
- **Winter temperature inversions:** from late October through January, cold, dense air settles near the surface beneath a warmer air layer, suppressing vertical mixing. Combined with calm winds, this "inversion lid" traps locally emitted and transported pollutants close to ground level — precisely the period covered by this dataset.
- **Regional agricultural burning:** post-monsoon paddy-stubble burning in the neighbouring states of Punjab and Haryana injects large quantities of particulate matter into the regional atmosphere each autumn; prevailing northwesterly winds carry this smoke into the Delhi airshed, compounding local emissions even after the burning season tapers off.
- **Dense vehicular and combustion sources:** a large and growing vehicle fleet, diesel generators, solid-fuel use, and construction dust contribute continuously to primary emissions of PM, NO_x, CO, and SO₂ within the city itself.
- **Urban geography:** rapid, dense construction has reduced green cover and open space that would otherwise aid natural ventilation and pollutant deposition, while the flat terrain around Delhi offers no orographic assistance in breaking up stagnant air masses.

This report uses hourly measurements of eight pollutants — carbon monoxide (CO), nitric oxide (NO), nitrogen dioxide (NO₂), ozone (O₃), sulphur dioxide (SO₂), fine particulate matter (PM_{2.5}), coarse particulate matter (PM₁₀), and ammonia (NH₃) — to ground these known environmental challenges in the specific data-generating period of January 2023, a month that falls squarely within Delhi's peak inversion season.

2. Research Questions

The analysis is organised around four research questions:

- **RQ1 — Pollutant dominance:** Which pollutant(s) most frequently determine Delhi's composite AQI, and how severe is air quality over the study period relative to national and international standards?
- **RQ2 — Diurnal variation:** What time-of-day patterns exist in pollutant concentrations, and are they consistent with known meteorological mechanisms (boundary-layer mixing, traffic cycles, nocturnal inversion)?
- **RQ3 — Human-activity signal:** Do pollutant levels differ systematically between weekdays and weekends, indicating a traffic/commercial-activity contribution distinct from the regional pollution baseline?

- RQ4 — Pollutant relationships and sources: How are the measured pollutants correlated with one another, and what do these relationships suggest about shared versus distinct emission sources?

3. Data and Methodology

3.1 Dataset

The dataset comprises 561 hourly records from 1 January 2023 00:00 to 24 January 2023 08:00, with concentrations ($\mu\text{g}/\text{m}^3$) for CO, NO, NO₂, O₃, SO₂, PM_{2.5}, PM₁₀, and NH₃, and no missing values. Timestamps are recorded in UTC in the source file; for diurnal analysis they were converted to India Standard Time (UTC+5:30) so that time-of-day patterns reflect Delhi's actual local clock rather than the recording timezone.

3.2 AQI Calculation

The composite Air Quality Index was calculated following the CPCB National AQI methodology. Each pollutant's concentration was first converted into an averaging window consistent with CPCB practice — a 24-hour rolling mean for PM_{2.5}, PM₁₀, NO₂, SO₂, and NH₃, and an 8-hour rolling mean for CO and O₃ — and then mapped to a 0–500 sub-index using the official piecewise-linear CPCB breakpoint tables. The overall AQI for each hour is the maximum of the individual pollutant sub-indices, and the pollutant producing that maximum is recorded as the "dominant pollutant" for that hour, per CPCB convention. The first 23 hours of the series were excluded while the rolling averages accumulated sufficient history, leaving 556 hourly AQI estimates.

CPCB defines six AQI categories: Good (0–50), Satisfactory (51–100), Moderate (101–200), Poor (201–300), Very Poor (301–400), and Severe (401–500).

3.3 Statistical Methods

- Descriptive statistics and CPCB-category classification of the derived AQI series.
- Diurnal profiling via hour-of-day (IST) averaging of PM_{2.5}, NO₂, and CO.
- Weekday–weekend comparison using Welch's two-sample t-test (unequal variances) on PM_{2.5}, NO₂, CO, and AQI.
- Pearson correlation matrix across all seven primary pollutants to assess shared-source signatures.

Because the dataset spans only 24 days within a single winter month, it captures short-term (daily/diurnal/weekly) dynamics with high fidelity but cannot, on its own, characterise Delhi's full annual seasonal cycle (e.g., the cleaner monsoon months). This constraint is revisited in Section 7 (Limitations).

4. Results

4.1 Overall Air Quality Levels (RQ1)

Across the 556 valid hourly observations, the mean composite AQI was 393 (SD 56), with a minimum of 106 and a maximum reaching the index ceiling of 500. Air quality never fell into the Good, Satisfactory, or Poor categories during the study window.

76% Severe hours (AQI 401-500)	22% Very Poor hours (301-400)	2% Moderate hours (101-200)	24 / 24 Days > WHO PM2.5 guideline
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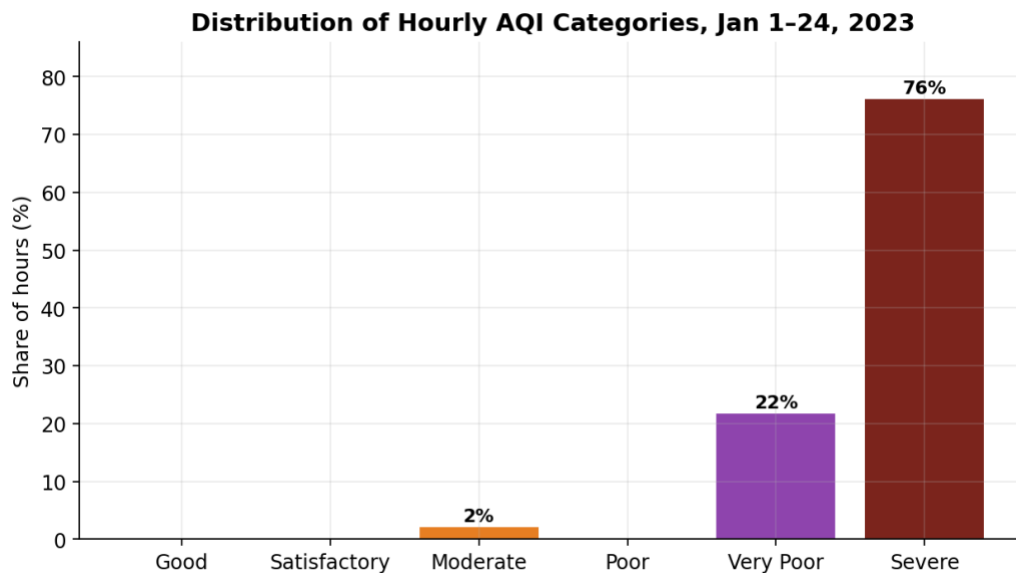


Figure 1. Distribution of hourly AQI categories, 1–24 January 2023. "Severe" and "Very Poor" together account for 98% of all hours observed.

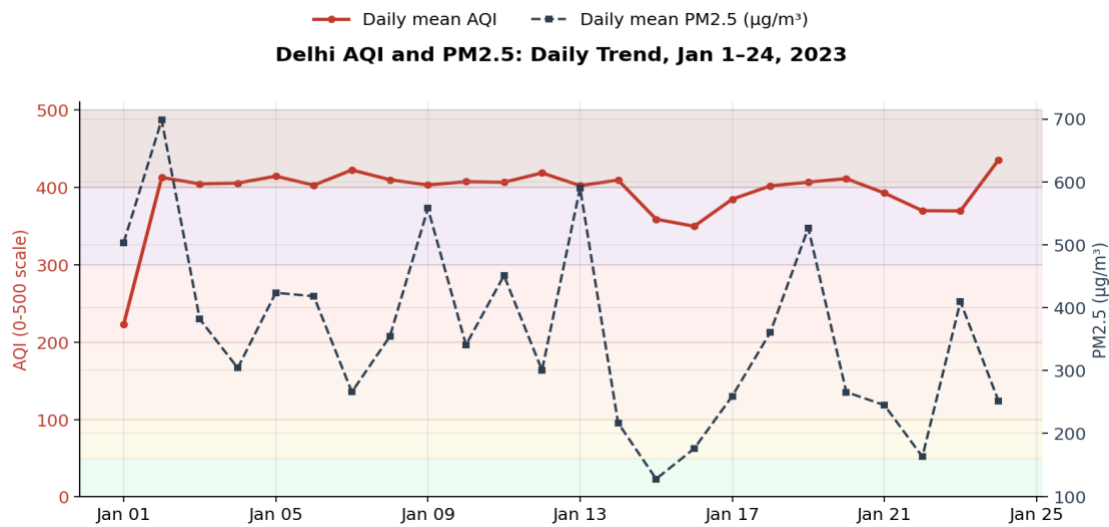


Figure 2. Daily mean AQI (left axis, shaded CPCB category bands) and daily mean PM2.5 (right axis) across the study period. Day-to-day swings of 200+ AQI points reflect changing wind and dispersion conditions rather than changing emissions alone.

Daily mean PM_{2.5} ranged from 128 µg/m³ (15 January, the cleanest day observed) to 698 µg/m³ (2 January, the most polluted day observed) — a more than five-fold swing within the same three-week window, underscoring how strongly short-term meteorology (wind speed, mixing depth) modulates pollution levels even when underlying emissions are comparatively stable.

PM_{2.5} was the AQI-determining ("dominant") pollutant in 90% of hours, consistent with fine particulate matter being the chronic, defining pollutant of Delhi's winter smog. CO was dominant in a further 6% of hours (concentrated during the evening traffic peak, see 4.2), with NO₂, SO₂, PM₁₀ and O₃ occasionally determining the index during brief windows.

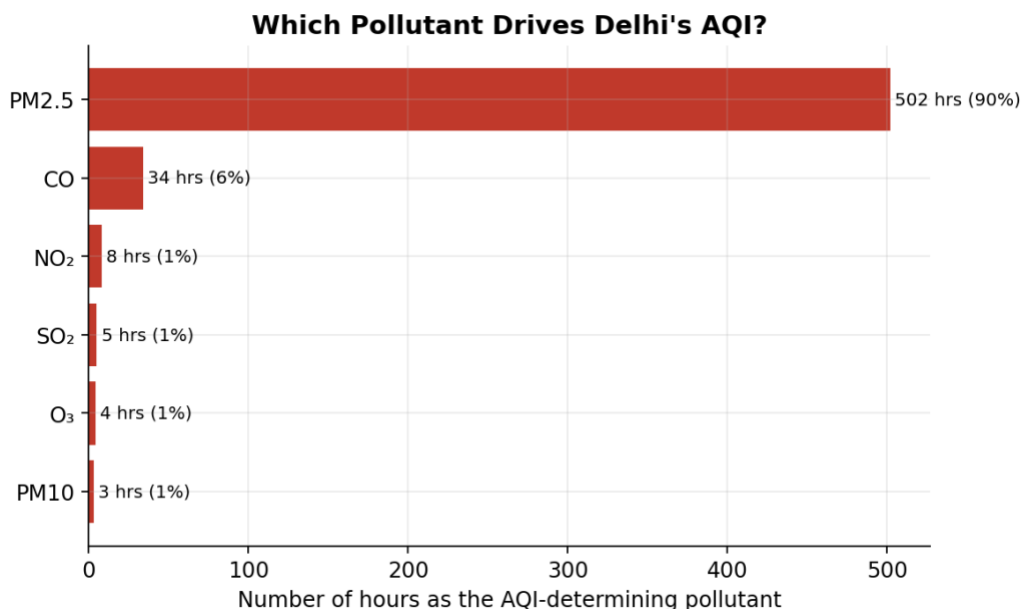


Figure 3. Frequency with which each pollutant's sub-index was the highest (and therefore AQI-determining) across the 556 hours analysed.

4.2 Diurnal Variation (RQ2)

Once timestamps are aligned to India Standard Time, pollutant concentrations show a clear and mechanistically consistent bimodal daily cycle. PM_{2.5} and NO₂ are elevated overnight and in the early morning, fall to their daily minimum in the early-to-mid afternoon (roughly 12:00–15:00 IST), and then climb sharply from late afternoon into a pronounced evening/night peak (approximately 19:00–23:00 IST).

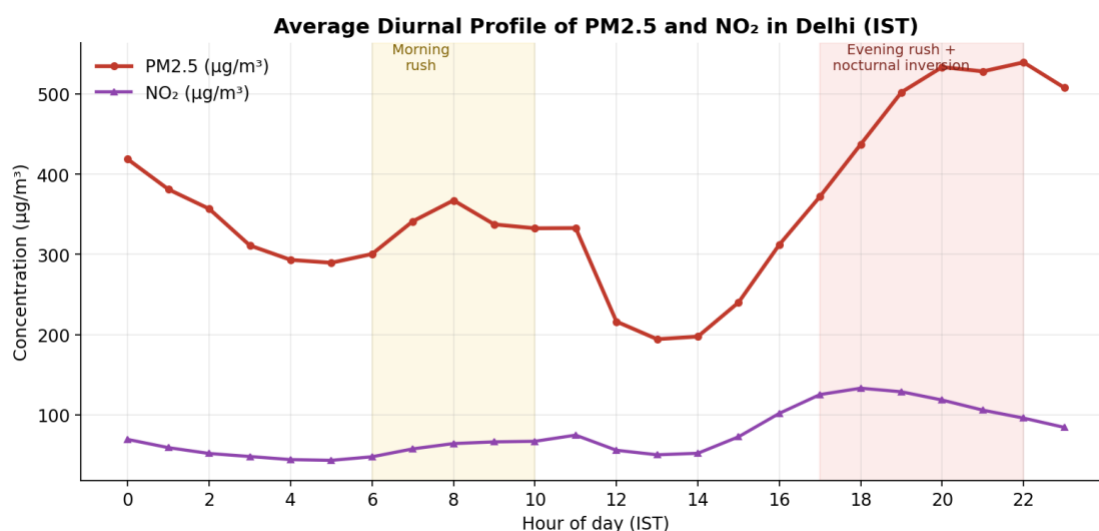


Figure 4. Average diurnal profile of PM_{2.5} and NO₂ by hour of day (IST). The midday dip coincides with maximum daytime convective mixing; the evening rise coincides with rush-hour traffic emissions layered onto the collapsing boundary layer as the surface cools after sunset.

This pattern is consistent with the combined effect of two mechanisms: (1) traffic and commercial activity, which raise primary emissions of NO₂ and CO through the morning and especially the evening rush hours, and (2) the diurnal cycle of the atmospheric boundary layer, which is deep and well-mixed by early afternoon (diluting pollutants) and shallow after sunset (concentrating them). NO₂ shows a sharper, more clearly traffic-timed evening peak (roughly 60% higher than its afternoon minimum) than PM_{2.5}, consistent with NO₂ being more directly tied to fresh vehicular exhaust while PM_{2.5} reflects a mix of fresh emissions and accumulated regional aerosol.

4.3 Weekday vs. Weekend Patterns (RQ3)

Mean concentrations were significantly higher on weekdays than weekends for every pollutant tested. PM_{2.5} averaged 401 µg/m³ on weekdays versus 261 µg/m³ on weekends (Welch's $t = 8.2$, $p < 0.001$); NO₂ averaged 83 versus 58 µg/m³ ($t = 6.8$, $p < 0.001$); and CO averaged 4,342 versus 2,585 µg/m³ ($t = 7.1$, $p < 0.001$). Composite AQI was also significantly higher on weekdays (401 vs. 374, $p < 0.001$), though the gap is smaller in index terms because both weekday and weekend AQI values are frequently pinned near the Severe ceiling.

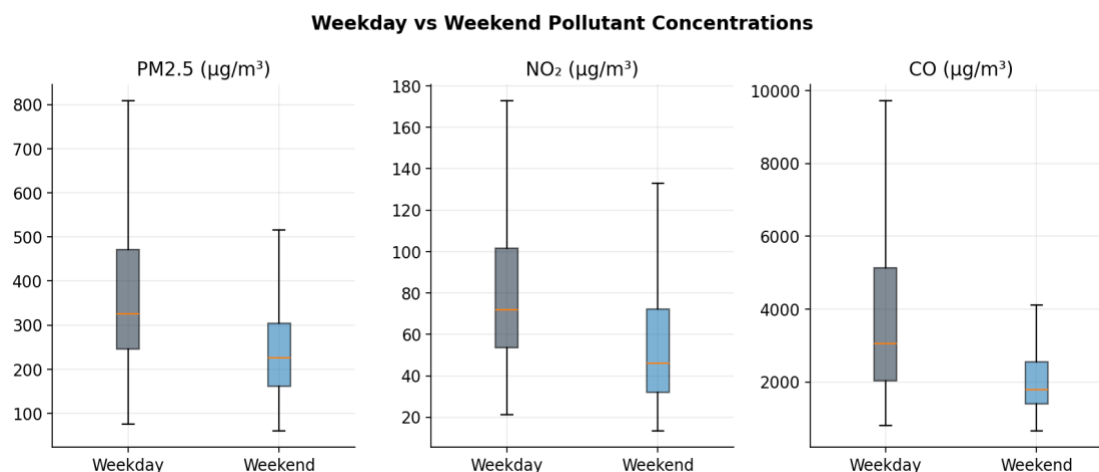


Figure 5. Weekday vs. weekend distributions of PM_{2.5}, NO₂, and CO (outliers hidden for clarity). Medians and interquartile ranges are visibly and significantly lower on weekends.

The magnitude of the weekday–weekend gap — roughly 35–50% higher pollutant levels on weekdays — indicates that vehicular traffic and commercial/industrial activity contribute a substantial, controllable share of Delhi's wintertime pollution on top of a persistent regional baseline that remains elevated even when human activity eases.

4.4 Pollutant Correlations and Source Signatures (RQ4)

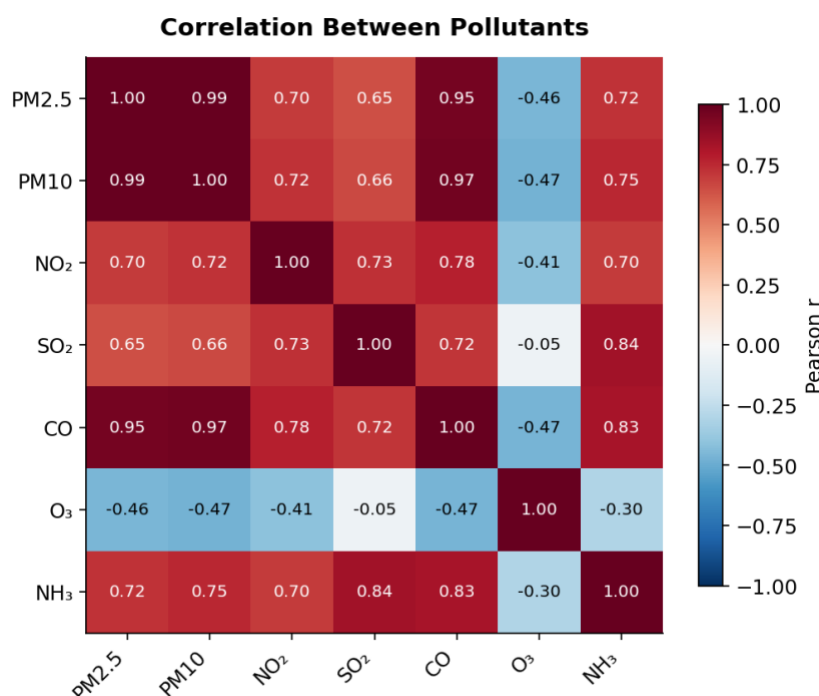


Figure 6. Pearson correlation matrix across the seven primary pollutants.

PM_{2.5}, PM₁₀, and CO are very strongly correlated with one another ($r = 0.95$ – 0.99), indicating they are largely driven by shared combustion sources — vehicular exhaust, biomass and solid-fuel burning, and industrial activity — and behave as a single "combustion-and-particulate" cluster under Delhi's winter meteorology. NO₂ and NH₃ correlate moderately-to-strongly with this cluster ($r = 0.70$ – 0.83), consistent with a shared traffic/combustion origin but with some independent variation, likely reflecting NH₃'s additional agricultural and waste-related sources. SO₂ correlates most strongly with NH₃ ($r = 0.84$), a signature often associated with coal combustion and secondary aerosol formation (ammonium sulphate). Ozone (O₃) is the only pollutant negatively correlated with the rest ($r = -0.30$ to -0.47): this is expected, since O₃ forms photochemically in sunlight and is simultaneously suppressed by high NO concentrations through titration, so it tends to be lowest exactly when primary combustion pollutants are highest.

5. Discussion: Linking Statistical Patterns to Geography and Meteorology

The statistical patterns above map closely onto Delhi's known environmental challenges. The near-permanent "Severe"/"Very Poor" AQI readings reflect the compounding of at least three factors operating simultaneously in January: (1) a shallow, stable winter boundary layer and frequent temperature inversions that physically trap emissions near ground level overnight and in the early morning; (2) continuing regional smoke transport and secondary aerosol formation carried into the Indo-Gangetic Plain by prevailing winds, layered on top of; (3) the city's own dense combustion emissions from traffic, industry, and residential heating/cooking.

The diurnal signal (Section 4.2) is essentially a daily microcosm of this same dynamic — the boundary layer 'breathes' in and out over 24 hours, deepening in the afternoon sun and collapsing after sunset, and pollutant concentrations rise and fall in near mirror-image response. The weekday–weekend gap (Section 4.3) shows that even within this meteorologically constrained system, locally controllable emissions (traffic, commercial activity) still make a large, statistically detectable difference — an encouraging sign for policy, since it means source control measures are not simply overwhelmed by regional/meteorological factors.

The correlation structure (Section 4.4) reinforces this: the particulate–CO–NO₂ cluster behaves as a single combustion signature, meaning that policies targeting vehicular and biomass combustion should move PM_{2.5}, PM₁₀, CO, and NO₂ together, while SO₂ and NH₃'s distinct co-variation suggests industrial/coal and agricultural sources need to be addressed through separate, complementary levers.

6. Public Health Implications

Sustained exposure at the levels recorded here — PM_{2.5} averaging roughly 24 times the WHO 24-hour guideline on every day of the study — carries well-documented health consequences, including exacerbation of asthma and chronic obstructive pulmonary disease, increased cardiovascular and cerebrovascular events, impaired lung development in children, and elevated risk of premature mortality with prolonged exposure. The concentration of the worst air quality in the overnight and early-morning hours (Section 4.2) is particularly relevant for outdoor workers, delivery and transport workers, and anyone exercising or commuting during this window, while the pronounced evening rush-hour peak affects the millions of commuters and roadside vendors active during that period.

This is a sensitive public-health topic; readers who are personally concerned about the health effects of air pollution exposure are encouraged to consult a physician or local public-health authority for guidance tailored to their circumstances.

7. Limitations

- Temporal coverage: the dataset spans only 24 days within January, entirely inside Delhi's peak winter inversion season. It cannot capture true seasonal variation (e.g., the comparatively cleaner monsoon or pre-monsoon periods); all statements about "seasonality" in this report should be read as characterising winter dynamics specifically, informed by the wider literature on Delhi's annual cycle rather than the data file itself.

- Single-site aggregation: the CPCB city-level AQI is normally computed from multiple monitoring stations; this dataset appears to represent a single location/aggregate feed, so intra-city spatial variation (e.g., roadside vs. background sites) is not captured.
- Averaging-window approximation: CPCB's official methodology specifies minimum data-completeness thresholds for 24-hour and 8-hour averages; the rolling-window approach used here approximates but does not exactly replicate the official reporting procedure.
- No co-located meteorological data: wind speed, boundary-layer height, and temperature were not available in this dataset, so the meteorological explanations offered in Section 5 are grounded in established atmospheric science and the published literature on Delhi's winter pollution, rather than measured directly alongside the pollutant data.

8. Recommendations

8.1 For Air Quality Management

- Prioritise fine particulate matter (PM_{2.5}) as the primary compliance metric, given it determines the AQI in the large majority of hours.
- Target the combustion cluster (PM_{2.5}/PM₁₀/CO/NO₂) jointly through vehicular emission standards, traffic demand management, and restrictions on open burning, since these pollutants share sources and move together.
- Address SO₂/NH₃ sources (industrial/coal combustion and agricultural/waste emissions) through separate, sector-specific interventions rather than assuming traffic policy alone will resolve them.

8.2 For Timing of Interventions

- Concentrate traffic-restriction and enforcement measures (e.g., odd-even vehicle schemes, staggered commercial hours) around the evening rush period, when both traffic emissions and a collapsing boundary layer combine to produce the sharpest pollution peaks.
- Issue public-health advisories that account for the overnight/early-morning pollution plateau, not only the visually obvious evening peak, since outdoor exposure risk remains high through the night.

8.3 For Public Health Response

- Direct at-risk groups (children, elderly, those with respiratory/cardiovascular conditions, outdoor workers) toward real-time AQI advisories, with specific guidance to limit outdoor exertion during the overnight-to-morning and evening peak windows identified in this analysis.
- Expand access to indoor air filtration in schools and healthcare facilities, given that "Severe" conditions were recorded in three-quarters of all hours studied.

9. Conclusion

Delhi's January 2023 air quality record illustrates, in a single winter month, why the city is regarded as one of the world's most air-pollution-burdened megacities. AQI in the Severe or Very Poor category for 98% of hours is not

an anomaly but the expected outcome of the interaction between a continental, inversion-prone winter climate, regional pollutant transport across the Indo-Gangetic Plain, and dense local combustion emissions from traffic and other sources. At the same time, the significant weekday–weekend gap and the sharp, traffic-aligned diurnal cycle show that locally controllable emissions still leave a measurable fingerprint on top of this regional baseline — meaning that well-targeted, well-timed policy interventions have real room to make a difference, even within a challenging geography and climate.

10. Appendix: Analytical Tools and Libraries

All data processing, statistical analysis, and chart generation for this report were performed in Python. The libraries used were:

Library	Role in this analysis
pandas	Loading and cleaning the CSV, timestamp handling and UTC→IST conversion, rolling-window averages, groupby aggregations (hourly/daily/weekday), resampling.
NumPy	Numerical operations underlying the AQI sub-index calculations and array handling.
SciPy (scipy.stats)	Welch's two-sample t-tests for the weekday vs. weekend comparisons (PM2.5, NO ₂ , CO, AQI).
Matplotlib	All charts: AQI/PM2.5 time series, diurnal profile, correlation heatmap, AQI category bar chart, dominant-pollutant bar chart, and weekday/weekend boxplots.

The Word document itself was assembled programmatically using the docx (Node.js/JavaScript) library, which embedded the Matplotlib-generated PNG charts alongside the formatted text, tables, and headings.